Inputs and source isolation

File: 01_inputs.kicad_sch

Reference low-pass crossover

File: 02_crossover.kicad_sch

Level and phase

File: 03_level_phase.kicad_sch

Subsonic, polarity and output

File: 04_protection_output.kicad_sch

Control and isolated trigger

File: 05_control.kicad_sch

Power and rail supervision

File: 06_power_mute.kicad_sch

Relay drivers and PSU interlock

File: 07_drivers.kicad_sch

HARDWARE RAIL-FAULT INTERLOCK

File: 08_rail_interlock.kicad_sch

REF L/R or LFE -> relay signal/return selection -> buffered mono -> REF LP / AV bypass
-> analogue level -> phase -> HP / AV bypass -> polarity -> output mute -> separate P68

No PCB layout. Separate P68 amplifier unchanged.

Sheet: /
File: SubController.kicad_sch

Title: SUBCONTROLLER — ANALOGUE SIGNAL PROCESSING

Size: A3	Date: 2026-10-08	**Rev: B**
KiCad E.D.A. 10.0.5		Id: 1/9

RCA shells MUST be insulated from plate/chassis. Inactive source has no bias / ESD path to board GND.

J1
REF LEFT
REF_L
REF_RETURN

J2
REF RIGHT
REF_R
REF_RETURN

J3
MARANTZ LFE
LFE
LFE_RETURN

K1A
AGN2004H
SEL_L
COM
NC
REF_L
NO
LFE

K1B
AGN2004H
SEL_R
COM
NC
REF_R
NO
LFE

K5A
AGN2004H
LIVE_L
COM
NC
NO
SEL_L

K5B
AGN2004H
LIVE_R
COM
NC
NO
SEL_R

NC = REFERENCE; energised = AV.
K1/K2 switch while K4 muted. Coils on driver sheet.

K5 OFF grounds buffer inputs; powered sources are disconnected.
K5 and K4 share RUN. Connect at CS code0, wait 2s, then fade up.

K2A
AGN2004H
K2A selects SOURCE_RETURN for differential input sensing. K2B unused.
Reference L/R returns share REF_RETURN before the relay.
Never bond SOURCE_RETURN to GND or chassis.
SOURCE_RETURN
COM
NC
REF_RETURN
NO
LFE_RETURN

TP8/9: selected REF L/R buffers; both carry LFE in AV.
Probe inactive RCA returns only with an isolated/differential instrument.

TP7
GND

K2B
AGN2004H
COM
NC
NO

LIVE_L
R1
47k / 0.1%
U1A
uPC4570G2
BUF_L
C1
100p C0G 50V
R2
47k / 0.1%
SOURCE_RETURN
R120
47k / 0.1%
R119
47k / 0.1%
C66
100p C0G / 50V

LIVE_R
R3
47k / 0.1%
U1B
uPC4570G2
BUF_R
C2
100p C0G 50V
R4
47k / 0.1%
SOURCE_RETURN
R122
47k / 0.1%
R121
47k / 0.1%
C65
100p C0G / 50V

U1 differential receivers: Vout = SIGNAL − SOURCE_RETURN (DC gain1).
47k matched ratios; REF input ~94k/channel, LFE ~47k. Return is sensed, not bonded.
C1/C64 and C2/C65: matched 100p C0G: differential RF pole ~33.9kHz.
0.1% resistors; select capacitor pairs within 2% for common−mode rejection.

BUF_L
R5
22K
BUF_R
R6
22K
U2A
uPC4570G2
SUM_INV
R7
10K

SUM_INV
R8
10K
U2B
uPC4570G2
MONO
R9
10K

MONO = (10/22)(L+R); AV = (20/22)LFE, after differential receivers.
No direct L/R connection. Main speakers remain full range.

TP8
BUF_L
TP9
BUF_R
TP10
MONO

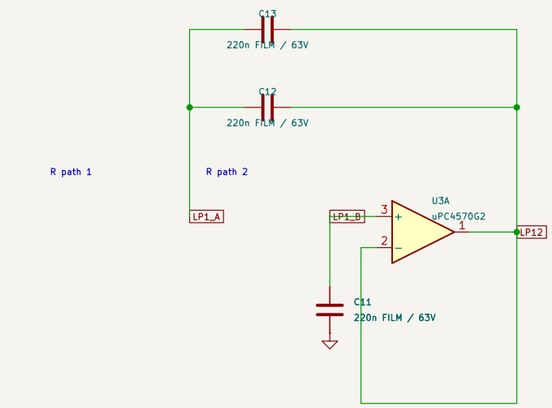
+15V
−15V
U1C
uPC4570G2
V+
V−
C3
100nF
C4
100nF
C5
1uF
C6
1uF
U1: caps physically at pins 8/4

U2C
uPC4570G2
C7
100nF
C8
100nF
C9
1uF
C10
1uF
U2: caps physically at pins 8/4

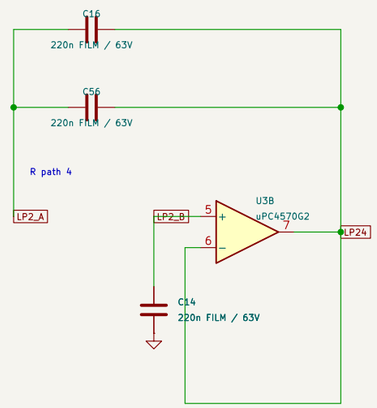
Audio remains analogue; no PCB layout or fabrication approval
DIY active subwoofer controller
Sheet: /Inputs and source isolation/
File: 01_inputs.kicad_sch
Title: Inputs and source isolation

Size: A3	Date: 2026−10−08	**Rev: B**
KiCad E.D.A. 10.0.5		Id: 2/9

Unity SK: Cf=440n (2x220n), Cg=220n; matched total R1/R2 => Q=0.7071.
RV1: one PCB-mounted PTD904-2015K-B203, four mechanically linked 20k linear gangs.

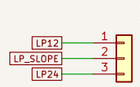


R1/R2 paths continue through RV1 gangs below

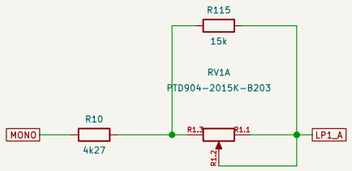


R1/R2 paths continue through RV1 gangs below

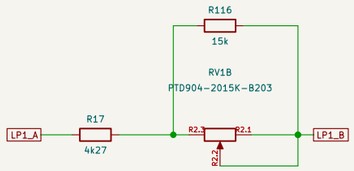
J4
SLOPE: 1-2=12 / 2-3=24



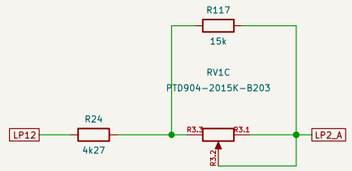
R115: reduces impedance / gang error



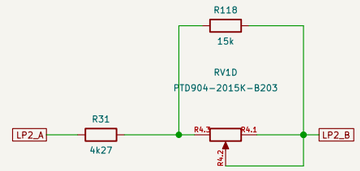
R116: reduces impedance / gang error



R117: reduces impedance / gang error



R118: reduces impedance / gang error



Gang R1: MONO -> LP1_A

Gang R2: LP1_A -> LP1_B

Gang R3: LP12 -> LP2_A

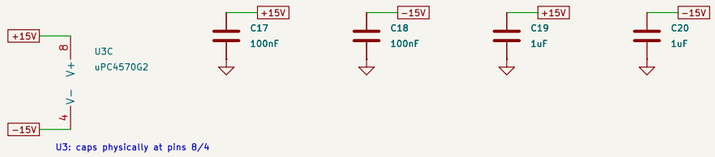
Gang R4: LP2_A -> LP2_B

R = 4k27 + (pot || 15k); Cf=440n, Cg=220n. Nominal 39.8..119.8Hz.
At 70Hz each physical rheostat is ~3k81. Max DC source resistance <28k.
Measure all four gangs and film caps; calibrate actual knob scale. Wiper tied to terminal1.
Keep RV1 <=50 C ambient. CW increases crossover frequency.

J25: 1-2 normal; 2-3 bench bypass. Change muted.

J25
LP: FILTER / BYPASS





Audio remains analogue; no PCB layout or fabrication approval

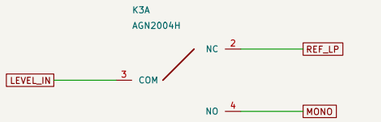
DIY active subwoofer controller

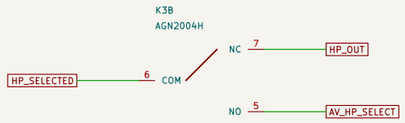
Sheet: /Reference low-pass crossover/
File: 02_crossover.kicad_sch

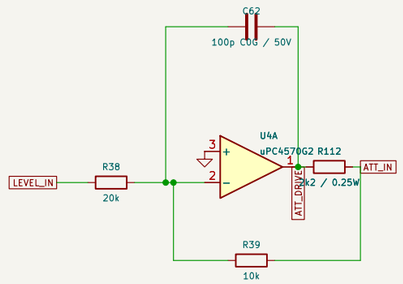
Title: Reference low-pass crossover

Size: A3	Date: 2026-10-08	**Rev: B**
KiCad E.D.A. 10.0.5		Id: 3/9

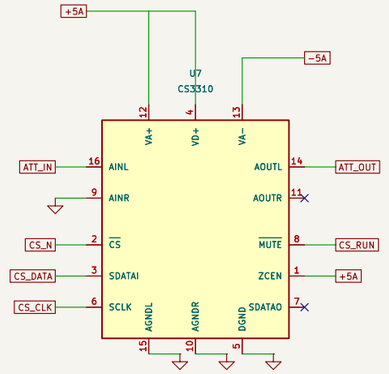
K3 pole A bypasses ONLY the Reference low-pass in AV. Pole B selects AV HP configuration on next sheet.





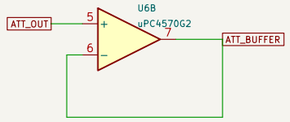


DC/audio gain −0.5; R112 limits CS input fault current <7.2mA.
R39 senses ATT_IN after R112; C62 provides local HF feedback.
HF corner ~110.5kHz (10k CS load); output −2 restores level.

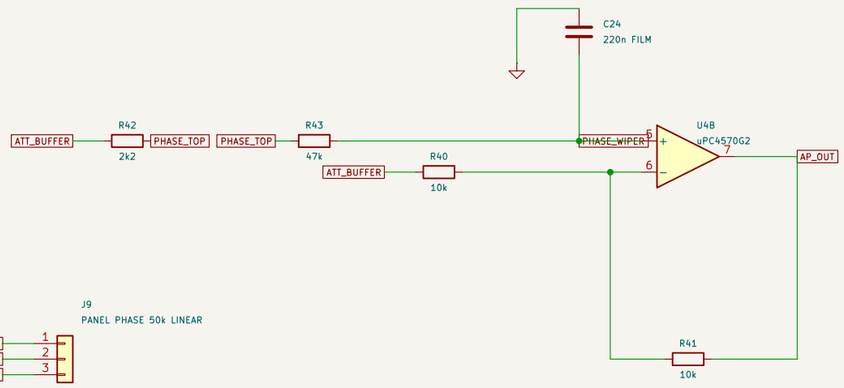


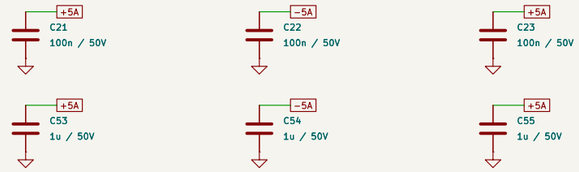
GAIN TUNE: R38=20k normal (input design ceiling 4Vrms).
R38=10k gives +6dB; use only with <=2Vrms sources and recalibrate both modes.

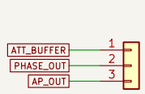
Left channel only. Right input grounded; right gain code=0 (mute).
REF and AV levels stored independently by MCU.
Firmware must prohibit all positive gain codes.

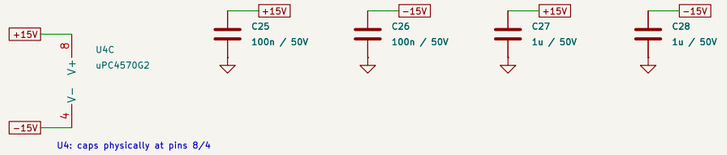


U6B isolates CS output from the HP / all-pass network.
CS sees only op-amp input / short trace capacitance (<100pF).









U4: caps physically at pins 8/4

Panel pot: pin1 top; pin2 wiper; pin3 tied wiper (fail-safe full R).
At 70 Hz: −24 to −137 deg; use bypass for 0 deg. Not constant time delay.

Audio remains analogue; no PCB layout or fabrication approval

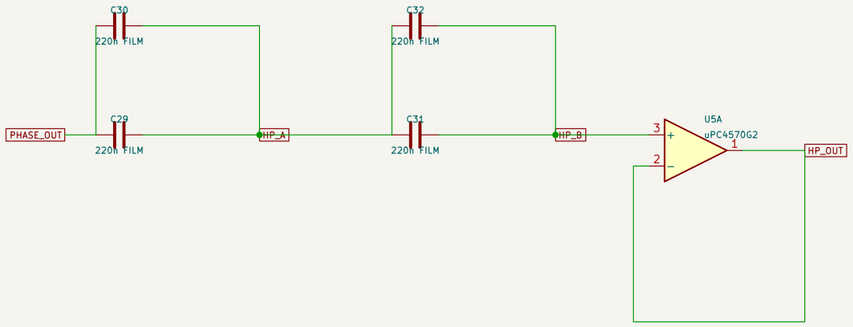
DIY active subwoofer controller

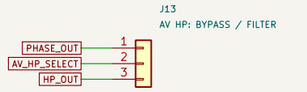
Sheet: /Level and phase/
File: 03_level_phase.kicad_sch

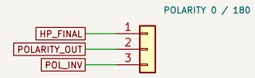
Title: Level and phase

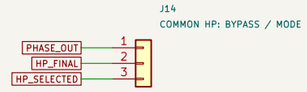
Size: A3	Date: 2026-10-08	**Rev: B**
KiCad E.D.A. 10.0.5		Id: 4/9

Default REF HP=20 Hz; AV HP bypass preserves AVR response. AV may select HP_OUT when protection required.

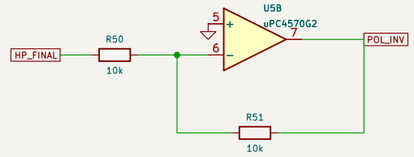


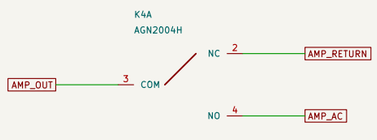


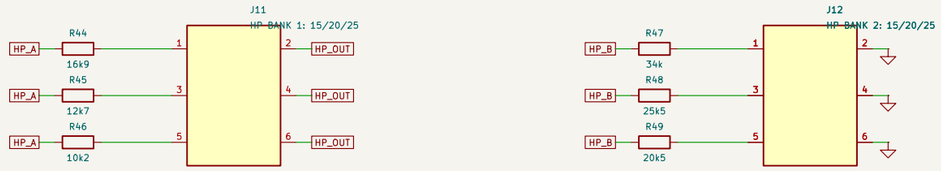




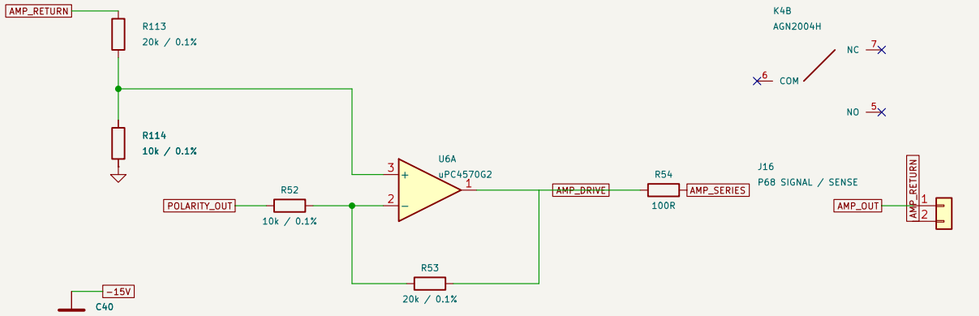
R1 bank: HP_A to HP_OUT. R2 bank: HP_B to GND.
C1=C2=440n; Q approx 0.707; unity gain above HP corner.

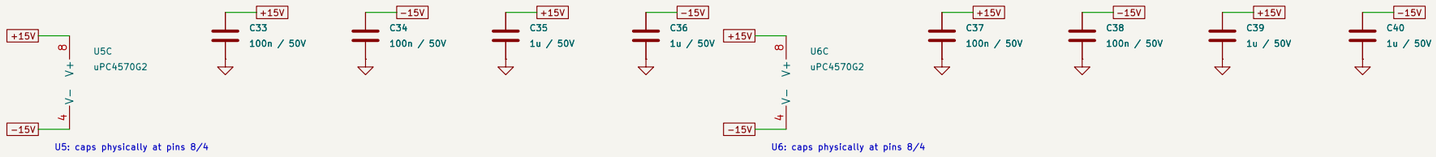






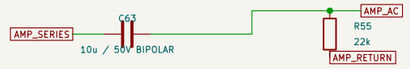
K4 OFF: P68 input shorted to its own SGND (AMP_RETURN).
U6A output = AMP_RETURN − 2*POLARITY_OUT; use 0.1% matched ratios.
J16.1 -> P68 J6.2 SIGNAL; J16.2 -> P68 J6.1 SGND (sense only).
Shared PSU 0V connects to P68 power star; no direct AMP_RETURN-to-GND bond.





U5: caps physically at pins 8/4

U6: caps physically at pins 8/4



C63: Nichicon UES1H100MPM, non-polar. R55 charges it while muted.
Keep mute >=2s after rail stabilisation/mode change; verify transient at J16.

Audio remains analogue; no PCB layout or fabrication approval

DIY active subwoofer controller

Sheet: /Subsonic, polarity and output/
File: 04_protection_output.kicad_sch

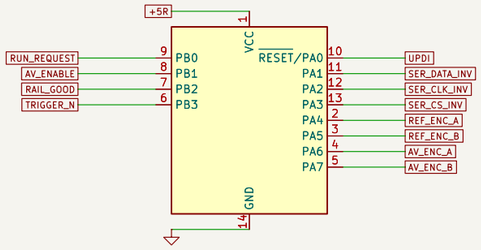
Title: Subsonic, polarity and output

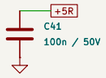
Size: A3	Date: 2026-10-08	**Rev: B**
KiCad E.D.A. 10.0.5		Id: 5/9

ATtiny1614 DIRECT CONTROL / RESET-SAFE MUTE / ISOLATED TRIGGER

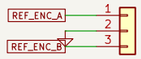
All audio stays analogue. MCU reset makes RUN and AV GPIO high impedance; external pull-downs drop relays.
Internal clock; BOD and watchdog required. PA0 remains UPDI. No I/O expander or retained relay latch.

U8
ATtiny1614-SS

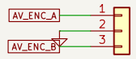




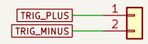
J18
REF LEVEL

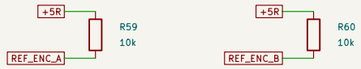


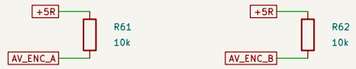
J19
LFE LEVEL



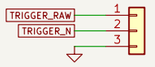
J20
ISOLATED TRIGGER 12V





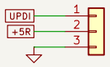


J24
AUTO / REF(open) / AV

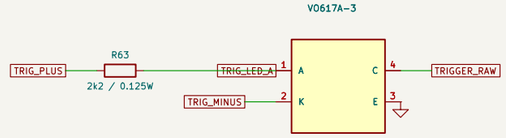


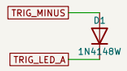
Dry-contact encoders: software quadrature debounce.
Separate stored REF/LFE levels: never positive CS gain.

J17
UPDI / +5R / GND



U10
V0617A-3





Open/inactive=REF, active=AV. J24: 1-2 AUTO, open REF, 2-3 AV.
12V trigger ~5mA; for 5V fit 750R. Trigger return stays isolated.



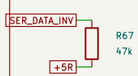


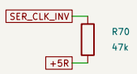


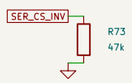












Serial collectors use +5A. 10kHz software clock, inverted GPIO. Default clock/data LOW, CS HIGH.

Audio remains analogue; no PCB layout or fabrication approval
DIY active subwoofer controller
Sheet: /Control and isolated trigger/
File: 05_control.kicad_sch
Title: Control and isolated trigger

Size: A3	Date: 2026-10-08	**Rev: B**
KiCad E.D.A. 10.0.5		Id: 6/9

SELV ONLY: externally regulated +/−15V input, NEVER P68 high-voltage power rails.
Nominal +/−15V; maximum +/−15.75V. Supply rating >=350mA positive / >=100mA negative.

+15V → D2 1N4007 → +5A
−5A → D3 1N4007 → −15V

J21
+15V / GND / −15V
+15V
−15V

C44
47u / 25V FR
+15V

C46
10u / 50V X5R
+15V

C45
47u / 25V FR
−15V

C57
10u / 50V X5R
+15V

C48
10u / 50V X5R
−15V

C58
10u / 50V X5R
−15V

PWR_FLAG +15V
PWR_FLAG −15V
PWR_FLAG

U11
TPS7A3901
1 INP
2 EN
3 NR/SS
5 INN
10 OUTP
9 FBP
8 BUF
7 FBN
6 OUTN
4 GND
11 EP

+15V
−15V
C59
100n / 50V

+5A
C60
10n C0G / 50V
R105
22k / 0.5%
R106
10k / 0.1%
R107
10k / 0.1%
R111
10k / 0.1%
R108
22k / 0.5%
R109
10k / 0.1%
R110
10k / 0.1%
−5A
C61
10n C0G / 50V

C47
10u / 50V X5R
+5A

+15V → D4 1N4007 → +5R

U13
7805 (Diodes stock)
+15V
IN
GND
OUT
+5R

C50
1u / 50V
+15V

C49
10u / 50V X5R
−5A

C51
47u / 25V FR
+5R

U11: budget 0.35W; solder EP to GND copper / thermal vias at layout.
U13: AV budget 2.7W; effective junction-to-ambient <=20 C/W, heatsink required.
C46+C57 / C48+C58 directly at IN pins; C47/C49 at OUT pins.
NR/SS100n; CFF10n each; keep CS VD+ tied to VA+. Local GND shares P68 power 0V.

TPS feedback: + arm (22k+10k)/10k -> +4.990V; − arm (22k+10k+10k)/10k -> −4.998V.
Negative lower arm returns to BUF (pin8), NOT GND. EN tied to +15V; EP11 tied to GND.

+5R → R74 680R → VREF_2V5

VREF_2V5 → R75 10k / 0.1% → VREF_1V25
R76 10k / 0.1%

U14
LM4040A25IDBZR
VREF_2V5

+15V → R83 100k / 1% → SENSE_POS15
R84 10k / 1%

VREF_2V5 → R85 15k / 0.1% → SENSE_NEG15
R86 180k / 0.1% → −15V

+5A → R87 24k / 1% → SENSE_POS5
R88 10k / 1%

VREF_2V5 → R89 20k / 0.1% → SENSE_NEG5
R90 86k6 / 0.1% → −5A

U15E
LM339
V+ 3
V− 12
+5R

C52
100n / 50V
+5R

D7
BAT54A-7-F
SENSE_NEG15 → 1
2 → SENSE_NEG5
3

U15A
LM339
SENSE_POS15 → 5
VREF_1V25 → 4
2 → RAIL_GOOD

U15B
LM339
VREF_1V25 → 7
SENSE_NEG15 → 6
1 → RAIL_GOOD

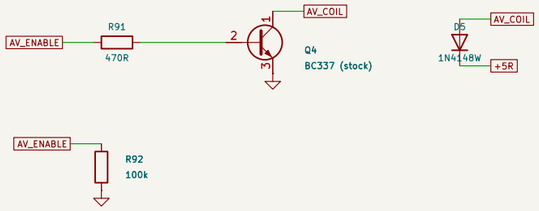
U15C
LM339
SENSE_POS5 → 11
VREF_1V25 → 10
13 → RAIL_GOOD

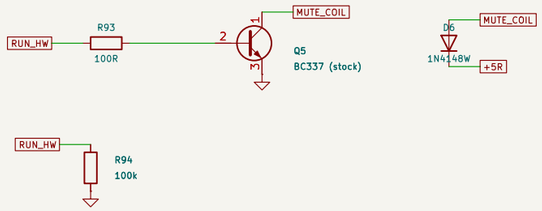
U15D
LM339
VREF_1V25 → 9
SENSE_NEG5 → 8
14 → RAIL_GOOD

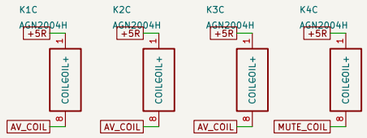
Nominal trip magnitudes: 13.75V / 13.75V / 4.25V / 4.1625V.
Negative sense ratios unchanged: D7 dual diode clamp asymmetric shutdown.
+/−5V monitors detect collapse, not full CS supply compliance. Require 500ms stable rearm.

Audio remains analogue; no PCB layout or fabrication approval
DIY active subwoofer controller
Sheet: /Power and rail supervision/
File: 06_power_mute.kicad_sch
Title: Power and rail supervision
Size: A3 | Date: 2026-10-08 | **Rev: B**
KiCad E.D.A. 10.0.5 | Id: 7/9

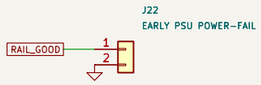
Default: AV_ENABLE=0, RUN_REQUEST=0. Output stays muted until firmware sequence completed.
Rail interlock clamps RUN_HW independently; MCU requires fresh settling delay after fault.





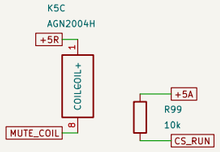


Q4 drives K1..K3: ~101mA nominal; allow 120mA. Q5 drives K4/K5: ~68mA; allow 80mA.
All AGN2004H coils intentionally use +5R (4.5V nominal).
R95 retains 1k fault-current limit; R93=100R increases Q5 base drive.
U13: allow 2.7W: effective junction-to-ambient <=20 C/W. +15V source >=350mA.



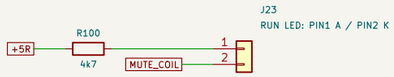




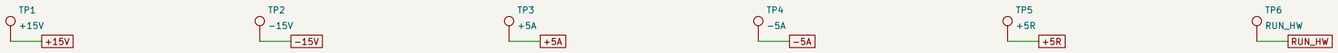




Optional open collector contact to GND. Assert before analog rails collapse.
NO external voltage injection. Without early warning, arbitrary fast
power loss cannot be guaranteed pop-free by a mechanical relay.



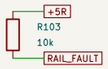
RUN=0: Q6 OFF, Q7 ON => CS mute asserted.
R97 uses +5A: loss of +5R still asserts mute while +5A lives.
RUN=1: Q6 ON, Q7 OFF => CS_RUN high; no rail backfeed.

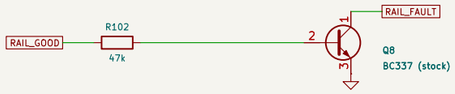


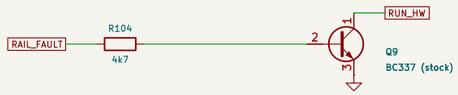
Audio remains analogue; no PCB layout or fabrication approval
DIY active subwoofer controller
Sheet: /Relay drivers and PSU interlock/
File: 07_drivers.kicad_sch
Title: Relay drivers and PSU interlock
Size: A3 | Date: 2026-10-08 | **Rev: B**
KiCad E.D.A. 10.0.5 | Id: 8/9

Rail comparators / external early power fail pull RAIL_GOOD low. ATtiny PB2 reads it independently of RUN_REQUEST.
On a fault: mute immediately in hardware; firmware clears RUN_REQUEST and waits 500ms stable RAIL_GOOD before rearm.









Good rails: Q8 ON, Q9 OFF; firmware may release RUN.
Bad rails: Q8 OFF, Q9 ON; RUN_HW clamped, Q5 OFF, K4 shorts input to AMP_RETURN.
+5R absent: all coils OFF; K4 shorts input to AMP_RETURN. No MCU instruction is required for this state.

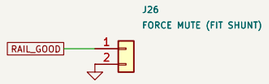
J26 shorts open-collector RAIL_GOOD to GND: hardware mute, independent of firmware.
TP11 post-LPF; TP12 post-phase; TP13 post-subsonic; TP14 final output.
Use GND TP7 as scope reference; never clip scope ground to an inactive RCA return.



















Audio remains analogue; no PCB layout or fabrication approval
DIY active subwoofer controller

Sheet: /HARDWARE RAIL-FAULT INTERLOCK/
File: 08_rail_interlock.kicad_sch

Title: HARDWARE RAIL-FAULT INTERLOCK

Size: A3	Date: 2026-10-08	**Rev: B**
KiCad E.D.A. 10.0.5		Id: 9/9